exactly the same manner when a bullet is fired into it. This reaction is analogous to the heating of a microscopic structure, if we bear in mind that an increase in the kinetic energy of the molecules in the sand pile corresponds to an increase in its temperature. In a nucleus, instead of molecules we deal with neutrons and protons, and the kinetic energy is associated with a certain temperature of the nucleus.

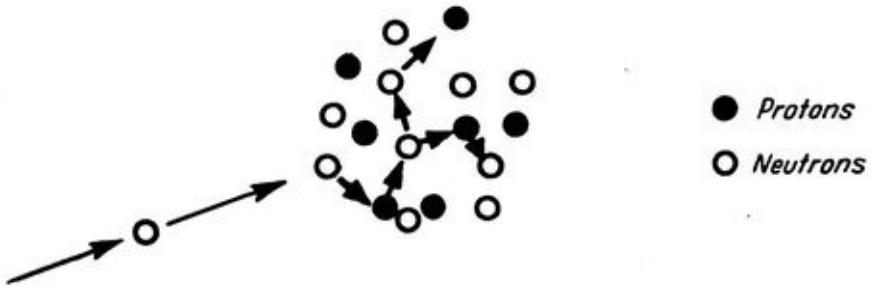


Figure 24.—Neutron penetrating a nucleus.

The temperature attained by a nucleus in this way—for instance, when a particle with an energy of approximately 8 Mev. is shot into it—has, in conformity with the laws of the kinetic theory of gases, the order of magnitude of ten thousand million degrees. It is roughly a thousand times higher than the highest temperatures that otherwise occur in the universe, the temperature in the interiors of the fixed stars. However, in this case, these high temperatures only affect the infinitesimally minute territory of that one nucleus.

If we imagine the nucleus in this state as a highly heated drop of liquid, the logically inevitable conclusion is that the nucleus will evaporate in consequence of the high temperature. This means that after a short while, some particle or another will emerge from the nucleus—generally that particle, the emergence of which is most advantageous from the point of view of energetics, in other words, the one the emergence of which requires the least energy. The energy necessary here corresponds to the heat of evaporation in an actual liquid. The nucleus cools off as a result of having yielded up this energy. Occasionally, a second particle will emerge, too. Or else, the residue of the energy, which is not sufficient for the emission of another particle, departs in the form of a gamma ray, as a photon. In a certain respect, this process is like the case of an incandescent drop of liquid which, due to its high temperature, emits visible light, too.

The process just described can continue unimpeded only if the particle used to bombard the nucleus is uncharged, i.e. if it is a neutron or a photon. But if the bombarding particle is a proton or an alpha particle, it must come up in the vicinity of the nucleus, against the potential barrier, the height of which is in proportion to the magnitude of the charge of the bombarded nucleus. The